

MASTER'S APPLIED PROJECT PROPOSAL

Physics-Based Table Tennis Robot Simulation and Control Validation in MuJoCo

Franka FR3 Arm with Analytical Policy Validation

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Abstract

This project will design, implement, and validate a high-fidelity MuJoCo physics simulation of a single Franka FR3 7-DOF robotic arm playing table tennis. Rather than training reinforcement learning (RL) policies (which is scoped as future work); this project will build and rigorously validate the simulation infrastructure and control pipeline that any future RL agent will rely on. The work comprises four deliverables: (1) a from-scratch MuJoCo simulation environment with physically accurate ball, table, net, and robot dynamics; (2) a paddle alignment test script that validates the precision and accuracy of the full action execution loop by commanding the arm to align its paddle face with an arbitrary vector in space; (3) an ideal policy emulation script that analytically computes the optimal impact position, paddle orientation, and paddle velocity from the ball's observed kinematic state and the simulation's physics model, then executes the intercept, demonstrating that the control pipeline can reliably return an incoming ball; and (4) a ball initial state validation suite that tests a range of spawn conditions to confirm physically correct bouncing behavior and unobstructed robot operation. Taken together, these deliverables produce a validated, RL-ready simulation platform as the foundation for future policy training.

1 Introduction and Motivation

Autonomous robotic manipulation in fast-paced, contact-rich environments remains one of the hardest open challenges in robotics. Table tennis distills many of these challenges into a single task: the robot must perceive a rapidly-moving ball, predict its future trajectory, plan a collision-free arm path, and strike the ball with a precisely-timed, correctly-oriented paddle face, all within a sub-second decision window.

Physics simulation has become the dominant training ground for such tasks. MuJoCo provides a differentiable, physically accurate simulation fast enough for model-free RL. This project will build a complete virtual table tennis arena: a single Franka FR3 7-DOF arm mounted at world frame position [1.5, 0, 0.76 m] on the robot's side of the table, with a custom paddle end-effector; a 40 mm hollow ball (2.7 g) spawned near the net ($x \approx 0$ m) traveling toward the robot with horizontal speed 1.2–2.0 m/s; and an ITTF-compliant table. The net is at $x = 0$ m; the opponent's half ($x < 0$ m) is the target landing zone. The robot must intercept the ball after it bounces on its own side and return it across the net.

The specific technical objectives of this project are:

- Construct a modular, config-driven MuJoCo simulation environment for a single-arm FR3 table tennis robot, including physically accurate ball, table, net, and robot dynamics.
- Develop a paddle alignment test script that validates the precision and accuracy of the action execution loop by commanding the arm to align its paddle face with an arbitrary target vector in world space and measuring position and orientation error.
- Implement an ideal policy emulation script that analytically computes the optimal impact position, paddle orientation, and paddle velocity from the ball's observed kinematic state and the simulation's physics model, then executes a full intercept to validate the end-to-end control pipeline.
- Write a ball initial state validation suite that tests a curated range of spawn positions, velocities, and spins to confirm physically correct bouncing behavior and unobstructed robot workspace, ensuring the environment is ready for future RL policy training.

This work will be conducted under the supervision of Dr. Nakul Gopalan in the Logos Lab at ASU, where it will also serve as the capstone Applied Project for the MS program in Robotics and Autonomous Systems (AI).

2 System Architecture

The system will be organized into three tightly-coupled layers, all physics constants and hyperparameters externalized into YAML configuration files.

2.1 Simulation Environment

The base simulation will be built on MuJoCo 3.x. The arena XML will define a single Franka FR3 7-DOF arm (fr3_link0 through fr3_link7, joint range: ± 2.9007 to $[0.44, 4.63]$ rad depending on joint) mounted at world frame $[1.5, 0, 0.76]$ m. A custom paddle end-effector will attach to fr3_link8: a 157 mm $\times$ 152 mm $\times$ 6.5 mm box geom (paddle_blade, mass 90 g, rubber friction 0.9) plus a cylindrical handle and an STL mesh visual. Two named sites, paddle_contact (position) and paddle_normal (orientation), will expose the blade's world-frame pose to the control pipeline. The environment wrapper will handle episode resets, ball spawning from a curated set of validated initial states, and termination detection (ball-hits-ground, ball-out-of-bounds, or episode timeout).

2.2 Ball Trajectory Predictor

A lightweight analytical predictor, also embedded in the control pipeline, will integrate the ball's observed kinematic state forward under gravity with a configurable empirical bounce coefficient-of-restitution ($COR = 0.48$). The predictor will search for the earliest post-bounce intercept within the robot workspace (x: 0.78–1.80 m, y: ± 0.65 m, z: 0.78–1.72 m) across up to 4 post-bounce arcs, preferring the parabola apex for maximum reaction time. It will return a 4-tuple contract: (impact_position, impact_time, impact_velocity, bounce_flag). The impact velocity vector will directly drive the desired paddle-face orientation.

2.3 Inverse Kinematics and Trajectory Pipeline

The control pipeline will derive the target paddle-contact site position by offsetting the impact point by ball_radius (20 mm) + blade_half_thickness (3.25 mm) + 2 mm margin along the desired normal. The desired normal will be the negated, normalized impact velocity. Two IK solvers will be supported:

- NumericalIKSolver: will use SLSQP optimization minimizing weighted position + normal-alignment error, with soft joint-limit and self-collision penalties; convergence at 5 mm.
- JacobianIKSolver: will use iterative pseudo-inverse Jacobian updates ($\alpha = 0.1$, max 500 iterations, tolerance 5 mm).

Four trajectory planners will generate smooth joint-position command sequences at 1 kHz simulation frequency: MinimumJerkTrajectory (5th-order polynomial, default), CubicSplineTrajectory (C^2 piecewise cubic, scipy), BSplineTrajectory (degree-3 approximating), and TrapezoidalVelocityTrajectory (time-optimal bang-coast-bang). A kinematic feasibility check will enforce a minimum trajectory duration based on maximum safe joint speed (1 rad/s) to prevent commanded velocity violations.

3 Methodology and Validation

Development will follow a staged validation protocol. Visual debugging will be built into every diagnostic script.

Stage	Activity	Script	Acceptance Criterion
1	Config + env wiring	Environment setup script	Home pose loaded; base pos matches config
2	Paddle alignment sweep	Paddle alignment test	Pos err < 5 mm; normal err < 5°
3	Ball initial state validation	Ball spawn and bounce validation test	All tested states bounce correctly; robot workspace unobstructed
4	Ideal policy emulation	Analytical intercept script	Ball successfully returned to opponent half in $\geq 80\%$ of tested states

4 Project Timeline

Six two-week milestones aligned with the ASU Applied Project calendar.

Milestone	Period	Deliverable	Status
M1	January 2026	Single-arm FR3 sim setup, YAML config wiring, home pose validation	Complete
M2	February 2026	Physics-based ball trajectory predictor, IK + trajectory pipeline, paddle alignment test script	Complete
M3	March 2026	Ball initial state validation suite; ideal policy emulation script; 3D viewer overlays	Complete
M4	March 2026	End-to-end integration testing; acceptance criteria verification across all four deliverables	Complete
M5	April 2026	End-to-end integration testing; results documentation and code clean-up	In Progress
M6	April 2026	Final report submission and repository release	Planned

5 Expected Outcomes and Contributions

Successful completion will produce the following concrete contributions:

- **Open-source benchmark.** An open-source single-arm Franka FR3 MuJoCo table tennis simulation with modular YAML-driven configuration, which will serve as a reproducible benchmark for contact-rich robot learning.
- **IK + trajectory library.** A validated library of two IK solvers and four trajectory planners for a 7-DOF manipulator under sub-second timing constraints, applicable to any contact-rich manipulation task.
- **RL-ready platform.** A validated, RL-ready simulation platform with proven physics, control pipeline, and analytical oracle, forming the direct foundation for future RL policy training and sim-to-real transfer experiments detailed in Section 6.
- **Reward design template.** An ideal policy emulation methodology using physics-model-derived intercept computation, demonstrating analytically correct hitting behavior and validating the full IK and trajectory stack without requiring a learned policy.
- **Diagnostic methodology.** A suite of three validation scripts (paddle alignment test, ball initial state validation, and ideal policy emulation) with real-time 3D viewer overlays demonstrating best-practice staged sub-system validation.

Beyond the immediate deliverables, this project will build direct competency in contact-rich physics simulation, low-level robot control, and analytical planning, skills central to the research

agenda of the Logos Lab and directly applicable to physical AI in healthcare robotics, industrial automation, and collaborative human-robot interaction.

6 Future Work

The validated simulation platform delivered by this project is explicitly designed to enable the following future research directions:

- **RL policy training.** Train model-free RL policies (PPO, SAC, TD3, TQC, RecurrentPPO) directly in the validated MuJoCo environment using the established IK and trajectory pipeline. A comprehensive algorithm sweep across task-space and joint-space action modes will identify the best-performing configuration.
- **Reward shaping.** Design and iterate on a sparse event-driven reward model that incentivizes correct-face post-bounce hits, net crossing, and opponent-side landing, using the analytical oracle as a performance upper bound.
- **Sim-to-real transfer.** Deploy the best trained policy on a physical Franka FR3 robot, using domain randomization and system identification to close the sim-to-real gap.
- **Curriculum learning.** Progressively increase task difficulty by expanding ball spawn ranges, adding spin, and introducing opponent-side targeting objectives once a basic return policy is established.

7 Technical Stack

Component	Technology
Physics Simulation	MuJoCo 3.x (DeepMind), 1 kHz Newton solver
Robot	Single Franka FR3 7-DOF arm, base at [1.5, 0, 0.76] m world frame
End-Effector	Paddle blade (157×152×6.5 mm box) + cylindrical handle + STL mesh, 90 g, site: paddle_contact
Ball State Observation	Ball position, velocity, and spin (9-D) used by analytical predictor
IK Solvers	NumericalIKSolver (SLSQP, SciPy) + JacobianIKSolver (pseudo-inverse)
Trajectory Planners	MinimumJerk, CubicSpline, BSpline, TrapezoidalVelocity
Controller Options	PD, PID, ComputedTorque, Velocity, Impedance
Configuration	YAML configuration files
Language / Deps	Python3, NumPy, SciPy, MuJoCo, PyYAML
Version Control	Git, github.com/BhavyaShah1234/TableTennisMujoco (Apache 2.0)